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Graduation Project Sprint 0

<CrashLens>

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1 Introduction

Road traffic accidents represent a continuing challenge for transportation systems and insurance operations in the Kingdom of Saudi Arabia, as they frequently result in material vehicle damages that require accurate assessment before repair or insurance claim settlement. According to a national study on traffic accident analysis in Saudi Arabia, a total of 285,547 traffic accidents were recorded in the Kingdom during the year 1439H, reflecting the substantial volume of incidents that lead to vehicle damage and associated financial losses. These figures highlight the growing burden placed on damage assessment processes and insurance services within the country [1].

In Saudi Arabia, traffic accidents represent a major economic and administrative concern due to their impact on vehicle repair costs, insurance claims, and service efficiency. National studies emphasize that traffic accidents lead to significant material losses, including vehicle damage and infrastructure costs, which place pressure on insurance companies and official assessment authorities [2] [3]. As the number of vehicles continues to increase, traditional damage assessment methods—primarily based on manual inspections—are becoming increasingly time-consuming, resource-intensive, and prone to subjectivity [4].

Recent advances in artificial intelligence, particularly in computer vision and deep learning, have demonstrated strong potential in automating vehicle damage detection and severity estimation from images. Research in this field shows that AI-based systems can analyze vehicle images to identify damaged parts, classify damage severity, and support insurance-related decision-making with higher consistency and efficiency compared to purely manual approaches. Such technologies enable faster preliminary assessments, reduce operational delays, and support scalable insurance services.

From a national perspective, the adoption of intelligent digital solutions aligns with the objectives of Saudi Vision 2030, which emphasizes digital transformation, efficiency improvement, and the use of advanced technologies to enhance service quality across sectors, including transportation and insurance [5]. Implementing AI-powered damage assessment solutions contributes to improving operational efficiency, supporting data-driven decision-

making, and reducing reliance on manual processes.
This graduation project proposes CrashLens, an AI-powered mobile application that allows vehicle owners to submit images of damaged vehicles and receive an initial estimation of damage severity and approximate repair cost. The system aims to support users by providing a convenient and consistent preliminary assessment while offering insurance-related stakeholders structured and verifiable assessment reports. This document presents the problem addressed by the project, the proposed solution, product vision, roadmap, objectives, scope, required tools, and the Scrum team structure.

1.1 The Problem

Vehicle damage assessment after road accidents is currently a slow and largely manual process. In many cases, vehicle owners are required to visit physical inspection centers and wait for human assessors to evaluate the damage. This dependency on in-person inspection makes the process time-consuming and inefficient, particularly during peak periods when assessment centers must handle a high volume of insurance claims [4].

In addition to delays, manual vehicle damage assessment is inherently subjective. Different inspectors may provide varying evaluations for the same damage, resulting in inconsistent severity judgments and repair cost estimates. Such inconsistencies often lead to disputes between vehicle owners and insurance providers, reducing transparency and increasing the overall processing time of insurance claims.

For example, after a minor traffic accident, a vehicle owner may need to take time off work to visit an authorized inspection center, wait for an assessor to review the damage, and then wait again for an official cost estimate. In some cases, the assessment may differ from another evaluator’s judgment, leading to further delays, reassessments, or disputes with the insurance company.

Despite the widespread availability of digital images and advances in computing technologies, there is still limited adoption of intelligent systems capable of providing initial, data-driven damage assessments based on visual evidence. The absence of such systems represents a missed

opportunity to reduce processing time, improve assessment consistency, and enhance user experience. This project focuses specifically on addressing this gap by providing an automated, image-based approach for preliminary vehicle damage severity and cost estimation, supporting—but not replacing—official inspection procedures.

1.2 The Solution

To address the limitations of manual vehicle damage assessment, this project proposes **CrashLens**, an AI-powered mobile application that provides an initial vehicle damage evaluation using images uploaded by users. The system enables vehicle owners to capture and submit guided multi-angle photos (e.g., front, rear, side, and close-up shots of the damaged area), along with basic vehicle information such as model and year, through a user-friendly interface.

The application analyzes the submitted images using artificial intelligence to estimate damage severity and generate an approximate repair cost. At a high level, the system follows a structured machine learning pipeline that includes image preprocessing, automated severity estimation, and cost calculation based on vehicle-related inputs. This approach supports faster and more consistent assessments while reducing dependency on immediate physical inspection.

From a technical perspective, the solution leverages computer vision methods, where a convolutional neural network (CNN) is used to classify vehicle damage severity into predefined categories (minor, moderate, or severe). The predicted severity is then combined with vehicle information through a structured costing mechanism to produce an approximate repair cost estimate (e.g., cost ranges based on severity and vehicle attributes). Model refinement and performance optimization will be conducted iteratively throughout the project lifecycle.

Overall, the proposed solution aims to improve accessibility, reduce claim processing delays, and provide more consistent preliminary assessments compared to purely manual evaluations.

1.3 Product

1.3.1 Product Vision

For vehicle owners **who** need a fast and consistent initial vehicle damage assessment after an accident, **CrashLens** is an AI-powered mobile application **that** analyzes uploaded vehicle images to estimate damage severity and approximate repair cost. **Unlike** manual inspection methods that are slow and subjective, **CrashLens** provides a standardized, data-driven preliminary evaluation that supports early claim and repair decisions.

1.3.2 Product Roadmap

The development of the CrashLens system will follow an incremental agile approach using the Scrum methodology. The project will be implemented over two academic semesters, resulting in two major releases. Each release consists of multiple sprints, and each sprint delivers a usable increment of the system.

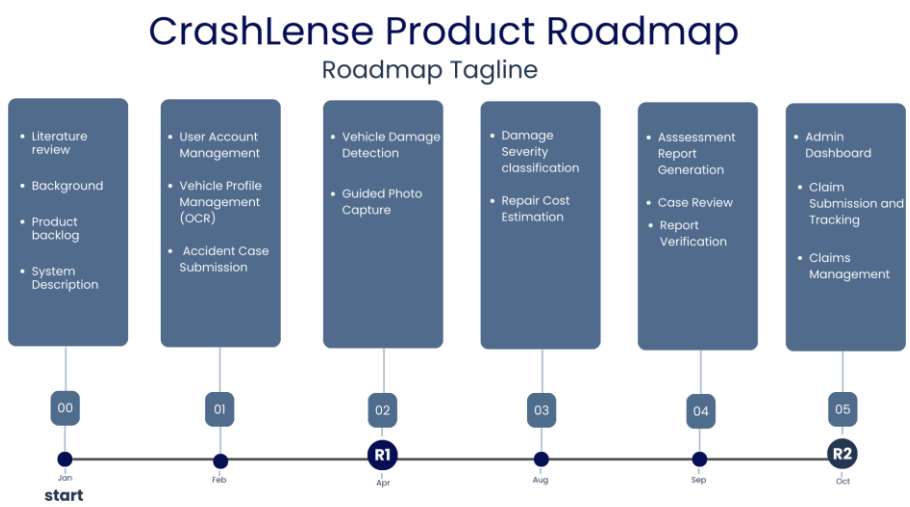


Figure 1: CrashLense Roadmap

1.3.3 Objectives

The objectives of this graduation project are defined on three levels: **product**, **project**, and **learning**.

1.3.3.1 Product Objectives (Customer focus – value)

The product objectives describe the value that CrashLens provides to its intended users, including vehicle owners and insurance-related stakeholders. By the end of the project, CrashLens aims to:

1. Provide a digital platform for vehicle owners to register and manage their accounts and vehicles, enabling structured and consistent use of the system.
2. Enable users to register accident cases and submit vehicle damage information, including accident details and related vehicle images.
3. Provide automated damage assessment capabilities, including damage detection, damage severity estimation, and repair cost estimation, to support preliminary evaluation of vehicle damage.
4. Generate automated damage assessment reports that summarize accident details, detected damage, severity estimation, and repair cost estimation.
5. Support report verification functionality to allow third parties to confirm the authenticity and status of generated assessment reports.
6. Provide an administrative dashboard that allows authorized administrators to review accident cases and approve or modify damage assessments.
7. Enable management of user objections, allowing users to submit objections to assessment results and administrators to review and respond to them through the system.

1.3.3.2 Project Objectives (Solution focus – plan)

the project objectives describe the technical and developmental activities required to design, implement, and deliver the CrashLens system across two academic semesters.

1. Conduct a background study and literature review to understand vehicle damage assessment processes and existing AI-based approaches.
2. Analyze and prepare publicly available datasets for vehicle damage assessment, including data cleaning, organization, and splitting.
3. Design the overall system architecture, including frontend, backend, database, and model interaction.
4. Design the machine learning workflow for vehicle damage assessment, including data preparation strategies and model selection at a conceptual level.
5. Develop the backend infrastructure and database structure required to store user, vehicle, accident, and assessment data.
6. Develop and train machine learning models for damage detection, damage severity estimation, and repair cost estimation.
7. Evaluate and improve model performance through testing, experimentation, and refinement.
8. Integrate the trained models into the system.
9. Perform system testing and refinement to ensure correctness, reliability, and usability.
10. Prepare final documentation and project deliverables.

1.3.3.3 Learning Objectives (Student focus – skills gained)

The learning objectives describe the knowledge and skills the project team aims to acquire beyond the standard IT curriculum through the development of the CrashLens system. By completing this project, the team will:

1. Gain practical experience in applying computer vision techniques to real-world image-based vehicle damage assessment problems.
2. Learn how to design and implement multiple machine learning models for related tasks, including damage detection, damage severity estimation, and repair cost estimation.

3. Develop hands-on experience with the complete machine learning workflow, including data collection, data preparation, preprocessing, augmentation, training, evaluation, and model refinement.
4. Learn practical transfer learning and fine-tuning techniques using convolutional neural networks (CNNs) for image-based classification and detection tasks.
5. Strengthen skills in backend and system integration, including connecting machine learning models with application services, databases, and user-facing components.
6. Gain experience in building and managing a data-driven software system using Agile/Scrum practices, including sprint planning, backlog management, teamwork, and iterative delivery across multiple releases.

1.3.4 Scope

This project focuses on the development of an AI-powered mobile application that provides a preliminary, image-based assessment of vehicle damage after road accidents. The system is intended for vehicle owners aged 18 and above, as well as insurance-related stakeholders who benefit from consistent and verifiable preliminary assessment reports.

The application allows users to create accounts, register vehicles, and submit damage reports through guided multi-angle photo capture and image upload. Based on the uploaded images and basic vehicle information, the system produces an initial damage severity classification (minor, moderate, or severe) along with an approximate repair cost estimate. Where feasible and supported by the selected datasets, the system may additionally highlight damaged regions or parts as an enhancement to improve result clarity.

The scope of the project is limited to exterior visible vehicle damage captured through images. The project supports the mobile platform only and the English language. Direct integration with official insurance company systems, issuance of legally binding claim decisions, and internal or mechanical vehicle diagnostics are explicitly outside the scope of this project

1.3.5 Hardware/Software Tools and Cost

Hardware Tools	
Name and Description	Cost
Laptops for development	Already available
Mobile for Testing	Already available
Software Tools	
Name and Description	Cost
Flutter **(mobile application development framework)**	Free
Figma **(UI/UX Design tool)**	Free
Firebase **(authentication, database, and storage)**	Free
Google Colab **(model development and experimentation)**	Free
Visual Studio Code **(VS Code) – IDE**	Free
OpenCV **(image processing and computer vision)**	Free
Programming **languages like (Dart, Python)**	Free
TensorFlow or PyTorch **(deep learning framework)**	Free
Kaggle datasets **(training and evaluation data)**	Free
Jira **(Scrum project management)**	Free
GitHub **(version control and collaboration)**	Free

Table 1: CrashLense Hardware/Software Tools and Cost Overvie

1.4 Scrum Team

1.4.1 Skill Set Requirements

Technical Skill Required	What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan
Mobile App Development (Flutter/Dart)	The team's level is basic; the team has basic knowledge in developing cross-platform mobile applications using Flutter. To strengthen proficiency, team members will complete targeted Flutter tutorials and implement core application screens during the first development sprint.
Backend (Python +Flask)	The team's level is basic, the team has introductory experience in backend. This skill will be developed further by implementing simple backend functions early in the project and gradually extending backend functionality across later sprints.
Database (Firebase)	The team's level is basic, the team has basic familiarity with Firebase services for authentication, database management, and cloud storage. The team will enhance this knowledge through practical implementation, starting with user authentication and image storage.
Machine Learning: Computer Vision + CNN Fine-Tuning	The team has beginner-level experience in machine learning and computer vision. During early sprints the focus will be on understanding the selected dataset and conducting baseline experiments. Later on, the team will develop and fine-tune pretrained CNN models using transfer learning for vehicle damage assessment tasks.

Data Preparation & Preprocessing (resize, normalize, augmentation)	The team has intermediate experience in preparing image data. This skill will be strengthened by applying consistent preprocessing pipelines and experimenting to improve model robustness.
Software Engineering Tools (Git/GitHub, Jira, versioning)	The team has intermediate experience in collaborative software development tools such as GitHub and Jira. These tools will be used throughout the project to manage sprint tasks, maintain version control, and support Scrum-based teamwork.

Table 2: CrashLense Team Key Skill Requirements.

1.4.2 Roles and Responsibilities

Scrum Team	
Product Owner:	Dr. Luluah Alhusain
Developers:	Atheer Budie Aryam Almutairi Leen binmueqal Sarah Alruwayte
Scrum Master (SM):	Dr. Luluah Alhusain
Stakeholders:	Saudi drivers Insurance companies

Table 3: CrashLense Team Roles And Responsibilities Overview

2 Background:

This chapter presents the domain and technical foundations required to understand the CrashLens project. It introduces vehicle damage assessment and explains the core technologies that enable automated, image-based damage evaluation. Since CrashLens relies on analyzing user-captured vehicle images, the background focuses on computer vision techniques for damage detection, severity estimation, and repair cost approximation, along with suitable performance metrics for evaluating system effectiveness.

2.1 The domain:

2.1.1 Vehicle Damage Assessment:

Vehicle damage assessment is a structured technical process conducted after road traffic accidents to determine the type, location, and severity of vehicle damage. The primary objective of this process is to identify the required repair actions and estimate the associated financial cost. Damage assessment represents a critical stage in post-accident procedures, as it forms the foundation for insurance claim settlement, repair planning, and compensation decisions [6].

The accuracy and reliability of a vehicle damage assessment depend on several factors, including the expertise of the assessor, the quality of available documentation, and adherence to standardized evaluation procedures. An incomplete or inconsistent assessment may lead to financial disputes between vehicle owners, repair workshops, and insurance providers. Therefore, damage assessment systems are typically governed by professional and regulatory standards to ensure fairness and transparency.

In Saudi Arabia, vehicle damage assessment activities are regulated under the supervision of accredited valuation authorities. Certified assessors operate through authorized centers that utilize unified electronic systems, such as Taqdeer [7], to issue standardized accident assessment reports. These reports serve as the official technical reference for determining repair feasibility and calculating compensation amounts.

2.1.2 Traditional Vehicle Damage Assessment Process:

The traditional damage assessment process is primarily manual and requires physical inspection of the vehicle. After an accident occurs, the vehicle owner must obtain an official accident report, such as a Najm report, and then visit an authorized assessment center. A certified assessor examines the damaged vehicle in person, documents the affected parts, captures photographic evidence, and evaluates the extent of the damage.

During this inspection, the assessor identifies damaged components, determines whether each part requires repair or replacement, and classifies the severity of the damage. The total repair cost typically consists of two main components: spare part cost and labor cost. Spare part cost depends on the vehicle's make, model, and whether original manufacturer parts or alternative parts are used. Labor cost reflects the time and technical effort required to repair or replace the damaged components.

Additional factors may influence the final estimate, such as paint type, regional labor rates, and workshop standards. For example, vehicles with metallic or specialized paint finishes may require additional materials and labor procedures, increasing the overall repair expense. Because these factors vary between workshops and regions, cost estimates are often expressed as a range rather than a fixed value.

Based on these findings, a structured assessment report is prepared, including technical descriptions and estimated repair costs. This report is then submitted to the relevant insurance company to initiate compensation procedures.

Although this process ensures compliance with regulatory standards and professional guidelines, it is time-consuming and dependent on human judgment. Scheduling physical inspections, documenting damage manually, and waiting for official approval may result in delays, particularly during periods of high accident volume. Furthermore, differences in assessor experience or workshop pricing practices may lead to variations in cost estimation.

2.1.3 Damage Classification Levels :

To maintain consistency and standardization, vehicle damages are typically categorized into defined severity levels. These levels guide repair decisions and influence the final cost estimation [8][9].

Minor damage generally refers to superficial or cosmetic impacts, such as small dents, light scratches, or minor bumper deformation. These damages often require limited repair procedures such as repainting, polishing, or minor part adjustments.

Moderate damage involves more significant structural or panel impact, including large dents in doors or fenders, partially broken bumpers, or noticeable deformation of exterior components. Such cases may require part replacement, welding, or more extensive repair work.

Severe damage represents critical structural or functional impairment. Examples include bent chassis components, damaged suspension systems, deployed airbags, or major frame distortion. Severe damage significantly increases repair complexity and cost, and in some cases may result in a total-loss evaluation.

Severity classification plays an essential role in determining the technical repair strategy and the financial value of the estimated damage.

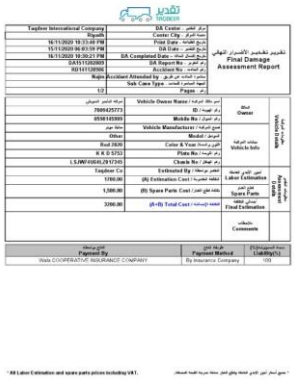
2.1.4 Damage Assessment Report:

Following inspection and cost evaluation, a structured damage assessment report is generated. Based on publicly available samples of official Taqdeer reports [8], such documents typically include the accident reference number, vehicle identification details, owner information, description of accident circumstances, list of damaged components, recommended repair or replacement actions, and an estimated cost breakdown. The report also contains assessor identification details and official validation to ensure authenticity.

The damage assessment report serves as crucial legal and financial function. Insurance companies rely on the report to determine compensation amounts and approve repair procedures.

Repair workshops use the document to understand the required scope of work. Vehicle owners depend on the report to review the assessment results and understand the estimated repair cost.

Once the report is approved, insurance compensation procedures are initiated according to the estimated value. In cases of disagreement, vehicle owners may submit objections or request reassessment, following regulated dispute resolution procedures.



تقدير
TAQDEER

Taqdeer International Company	DA Center / مركز التقدير	تقدير تقديم الأضرار النهائي Final Damage Assessment Report	
Riyadh	Center City / مدينة المركز		
16/11/2020 10:33:49 PM	Print Date / تاريخ الطباعة		
15/11/2020 06:03:59 PM	DA Date / تاريخ التقدير		
16/11/2020 10:30:21 PM	DA Completed Date / تاريخ إكمال الحالة		
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001411120006	Accident No / رقم الحادث		
Najm Accident Attended by / ساعد الحادث عن طريق			
	Sub Case Type / نوع الحادث الفرعي		
1/2	Pages / رقم		

مالك المركبة / Vehicle Owner Name / اسم مالك المركبة		المالك Owner	Customer Details تفاصيل العميل
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0550161000	Mobile No / رقم الجوال		
بلاط جوز	Vehicle Manufacturer / الصنع الشركة	بيانات المركبة Vehicle Info	Vehicle Details تفاصيل المركبة
Other	Model / الموديل		
Red 2020	Color & Year / اللون والسنة		
K K D 3752	Plate No / رقم اللوحة		
LSJW01051,(01124)	Chasis No / رقم الهيكل		
Taqdeer Co	Estimated By / التقدير من طرف	تقدير الأضرار التقديري Labor Estimation	Estimation Details تفاصيل التقدير
1700.00	(B) Estimation Cost / التكلفة التقديرية		
1,500.00	(B) Spare Parts Cost / تكلفة قطع الغيار	قطع الغيار Spare Parts	
3200.00	(A+B) Total Cost / التكلفة الإجمالية	المبلغ الإجمالي Final Estimation	
		ملاحظات Comments	

طرق الدفع Payment By	طريقة الدفع Payment Method	نسبة التحمل Liability(%)
Wala COOPERATIVE INSURANCE COMPANY	By Insurance Company	100

* All Labor Estimation and spare parts prices including VAT.

* جميع أسعار أجور الأضرار التقديرية وقطع الغيار متضمنة ضريبة القيمة المضافة.

Figure 2: tagdeer online public report sample[10]

2.2 Technologies:

This section presents the core artificial intelligence and computer vision technologies used in automated vehicle damage analysis. The discussion begins with foundational machine learning concepts, followed by deep learning architectures, detection and segmentation models, and concludes with task-specific performance evaluation metrics.

2.2.1 Machine Learning and Deep Learning

Machine learning is a branch of artificial intelligence that enables systems to learn patterns from data and make predictions without being explicitly programmed for every scenario. In visual analysis tasks, deep learning — a subset of machine learning — has become the dominant paradigm due to its ability to automatically extract complex features from raw image data[11].

Deep learning models rely on multi-layer neural networks that progressively learn hierarchical representations. In image-based tasks, early layers typically detect low-level features such as edges and textures, while deeper layers learn higher-level semantic patterns such as object parts and structures[12].

2.2.2 Transfer Learning and Fine-Tuning

Transfer learning is a training strategy where a model pre-trained on a large dataset is adapted to a new, domain-specific task. Since collecting large-scale labeled vehicle damage datasets is challenging, transfer learning significantly improves model performance and reduces training time [13].

Fine-tuning is a critical step in transfer learning. It involves:

- Retaining early convolutional layers that capture generic visual features.
- Re-training later layers to adapt to domain-specific patterns.

Fine-tuning improves generalization, reduces overfitting, and enhances model robustness in real-world conditions. Due to limited domain datasets in automotive damage analysis, fine-tuning is considered a crucial methodological component [14].

2.2.3 Computer Vision:

is a subfield of artificial intelligence (AI) that equips machines with the capability to process, analyze, and interpret visual inputs, such as digital images and videos. By utilizing machine learning, computer vision enables computers and systems to derive meaningful information from visual data. Within CrashLens, computer vision is used to interpret photographs of damaged vehicles captured via mobile devices [15][16].

2.2.4 Object Detection:

is a multifaceted computer vision task that functions by extracting both the semantic identity and the precise spatial coordinates of multiple entities within a single digital frame. At its core, the process is defined by the generation of bounding boxes, which are represented by a four-coordinate vector—typically (x, y, w, h) —defining the centre, width, and height of the detected

object. Each box is inextricably linked to a class label and a confidence score, a scalar value representing the model's architectural certainty that the identified region contains a valid object of interest.

In practical application, such as vehicle damage analysis, object detection serves as the primary diagnostic tool for localizing affected components. Rather than evaluating a scene globally, the model identifies specific regions of interest, such as a "fractured headlight" or "dent," by isolating those features from the background pixels. This allows for a granular analysis where a system can distinguish between a single point of impact and multiple, disparate areas of damage on a single vehicle, providing a mapped "hit-list" of required repairs.

The structural methodology for achieving this spatial intelligence is generally divided into two distinct architectural philosophies:

The Two-Stage Detector pipeline, exemplified by architectures like Faster R-CNN, operates through a "proposal-and-refinement" workflow. A Region Proposal Network (RPN) first identifies potential candidate areas that might contain objects, which are then passed to a second stage for final classification and bounding box regression. This decoupled approach is known for its high localization accuracy, particularly for small or overlapping objects.

In contrast, Single-Stage Detectors, such as the YOLO (You Only Look Once) family, treat detection as a singular regression problem. By dividing the image into a grid and predicting bounding boxes and class probabilities simultaneously in a single forward pass through the network, these models prioritize inference speed. This makes them the standard for real-time applications where rapid processing is as critical as spatial precision.

2.2.5 Image Segmentation:

represents the most granular tier of computer vision, transcending the spatial limitations of object detection. While detection identifies an object’s presence within a rectangular coordinate system, segmentation performs pixel-level classification, assigning a specific semantic label to every individual pixel in the image. This transition from "coarse localization" (boxes) to "fine-

grained partitioning" (masks) allows for a mathematical understanding of an object's exact geometry and contours.

In the context of vehicle damage analysis, this level of detail is transformative. By identifying the specific pixels that represent "deformed metal" or "shattered glass," segmentation models can calculate the exact surface area and boundaries of a dent. This provides a significantly more accurate metric for repair cost estimation than a bounding box, which inevitably includes "noise" pixels from the undamaged background surrounding the impact site.

The architectural methodology for achieving this pixel-perfect accuracy generally splits into two primary functional approaches:

- **Semantic Segmentation:** This approach treats the image as a collection of categories. Every pixel belonging to the "car" class is colored the same, regardless of how many individual cars are in the frame. It is essentially a dense classification task that focuses on the "what" and "where" across the entire scene.
- **Instance Segmentation:** This approach combines the strengths of object detection and semantic segmentation. It not only identifies the class of each pixel but also differentiates between separate objects of the same class. In a multi-car accident, instance segmentation would assign a unique ID to "Car A" and "Car B," ensuring their overlapping boundaries are clearly defined.

To process these complex spatial relationships, segmentation models typically utilize Encoder-Decoder architectures.

- **U-Net:** This symmetric architecture uses an "encoder" to capture the context of the image and a "decoder" to enable precise localization. It is famous for its "skip connections" that recover fine-grained spatial information lost during downsampling.
- **Mask R-CNN:** An evolutionary step above detection, Mask R-CNN adds a third branch to the standard Faster R-CNN architecture. While two branches predict the class and the bounding box, the third branch predicts a binary mask for each detected object, effectively merging detection and segmentation into a single high-performance pipeline.

2.2.6 Image Classification

serves as the foundational pillar of computer vision, tasked with assigning a definitive label to an entire image based on its global visual characteristics. While rudimentary classification identifies the subject of a photo, its application within the automotive sector has evolved into a sophisticated automated damage assessment pipeline. This end-to-end system begins with image acquisition, where multi-angle visual data is captured, followed by damage detection to isolate structural anomalies.

The core of this process is severity estimation, which utilizes classification algorithms to provide standardized evaluation outcomes—categorizing damage into discrete levels such as "Minor," "Moderate," or "Severe." This classification is critical because it informs the final stage of the pipeline: repair cost approximation, where the system translates visual severity into a data-driven financial estimate.

To handle the high variance in vehicle surfaces, such as reflections and complex geometries, the industry relies on advanced Deep Convolutional Neural Network (CNN) architectures:

- **ResNet**: Employs residual learning to overcome the degradation problem in very deep networks, ensuring stable feature extraction.
- **VGG**: Utilizes a simplified, deep architecture with small convolutional filters to capture fine-grained spatial hierarchies.
- **EfficientNet**: Optimizes performance by scaling depth, width, and resolution simultaneously, providing high-accuracy results with maximum computational efficiency.

Together, these models enable a seamless transition from raw visual input to precise, automated diagnostic reports.

2.2.7 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for image processing. CNNs consist of convolutional layers, pooling layers, and fully connected layers.

- Convolutional layers extract spatial features using learned filters.
- Pooling layers reduce spatial dimensions while preserving important features.
- Fully connected layers perform classification based on extracted representations.

CNNs form the backbone of most modern computer vision systems, including classification, detection, and segmentation tasks. Their ability to handle spatial hierarchies makes them particularly effective for analyzing complex real-world images.

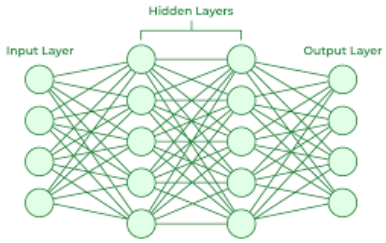

Figure 3: Neural network main structure.

2.2.8 YOLO (You Only Look Once)

YOLO is a single-stage object detection architecture designed for high-speed inference while maintaining competitive accuracy. Unlike two-stage detectors that first generate region proposals and then classify them, YOLO performs detection in a single forward pass through the network [17].

Key characteristics of YOLO include:

- Real-time detection capability.
- End-to-end training.
- Simultaneous prediction of bounding boxes and class probabilities.

YOLO variants (e.g., YOLOv5, YOLOv7, YOLOv8) introduce architectural improvements for better accuracy and computational efficiency. Due to its balance between performance and speed, YOLO has become widely adopted in applied computer vision systems.

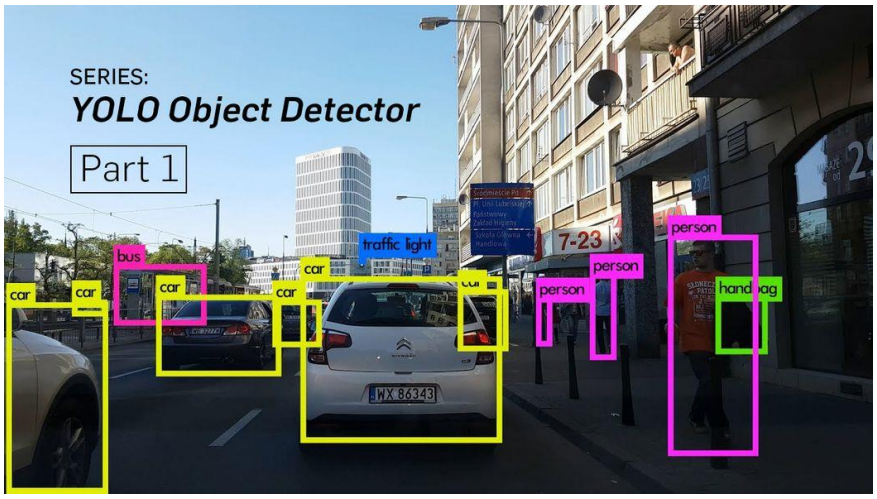

Figure 4:YOLO Multi-Object Detection and Classification

2.2.9 Performance Evaluation Metrics:

Standard evaluation metrics are used to assess the performance of the proposed computer vision models in detecting vehicle damage and affected regions. These metrics help measure detection accuracy, completeness, and localization quality, ensuring reliable and objective evaluation of the system [18][19].

True Positives (TP) :occur when the model correctly identifies and localizes an object of interest.

True Negatives (TN): occur when the model correctly identifies that an object of interest is not present.

False Positives (FP): occur when the model incorrectly identifies an object of interest where there is none.

False Negatives (FN): occur when the model fails to identify an object of interest that is present.

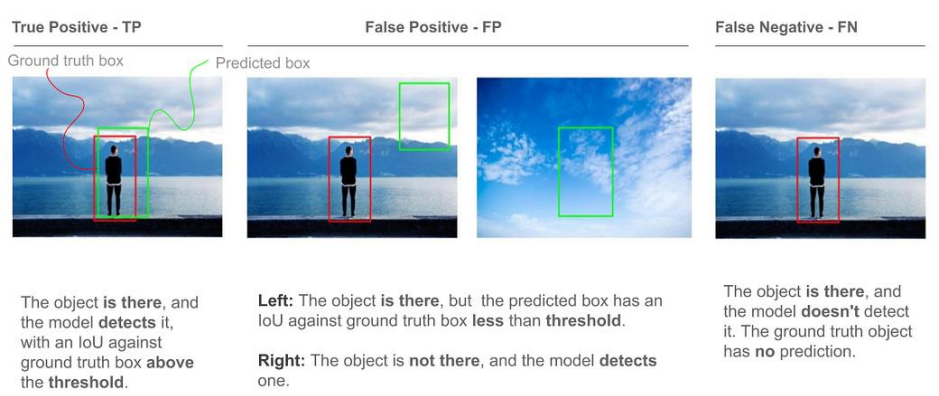

Figure 5: explaining tp,fp,fn

2.2.9.1 Detection Performance Metrics

Object detection models are evaluated based on both classification accuracy and localization quality.

- **Precision(P):**
  is a metric used to measure the accuracy of the positive predictions made by an object detection model.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

- **Recall:** is a metric used to measure the completeness of the object detection model in identifying all relevant objects within a dataset.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

  A high recall value means that the model successfully identifies most of the relevant objects, with few missed detections

- **Intersection over Union (IoU):**
  is a metric used to evaluate the accuracy of object detection models. It measures the overlap between the predicted bounding box and the ground truth bounding box. The IoU is defined as the ratio of the area of intersection to the area of union of the two bounding boxes.

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}}$$

- **Mean Average Precision (mAP):**
  is the meaning (average) of AP values across different classes and/or different IoU thresholds. so, the mAP is the meaning of the AP values across all classes in the dataset.

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} A\,P_i$$

  where N is the number of classes and AP_i is the average precision for the i-th class.

2.2.9.2 Severity Classification Performance Metrics

- **Accuracy**: evaluates how well a machine learning model performs. It represents the percentage of correct predictions made by the model.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision and Recall**

Used when class imbalance exists between severity levels.

- **F1 Score**:
  is a single metric that combines precision and recall providing a balanced measure of a model's performance.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1 Score ranges from 0 to 1, where 1 indicates perfect precision and recall, and 0 indicates the worst performance.

- **Confusion Matrix**

A confusion matrix provides detailed insight into misclassification patterns between severity categories, helping evaluate robustness and bias.

3 Literature Review

The integration of Artificial Intelligence (AI) and Computer Vision in vehicle damage assessment has been extensively explored to overcome the limitations of manual inspection. This chapter provides a structured review of existing literature, categorized into three core domains: damage detection, severity estimation, and repair cost estimation, focusing on the methodologies that form the foundation of this field.

3.1 Vehicle Damage Detection:

In the primary stage of identifying car damage, Zhang et al. (2020) [20] developed a specialized approach using an improved Mask R-CNN algorithm designed to detect small scale damages such as scratches and subtle dents. Their methodology involved refining the Feature Pyramid Network (FPN) to better capture multi scale features and implementing RoIAlign to ensure precise pixel-level edge detection instead of using standard bounding boxes. The model was trained on a self-collected dataset of 2,000 real-world accident images annotated with high-precision masks using the Labelme tool. The results demonstrated a detection accuracy of 96.68%, proving that precise segmentation is a critical requirement for accurate damage area calculation.

Furthermore, research by Jin et al. (2023) [21] explored the efficiency of the YOLOv8 architecture, which utilizes a single stage detection methodology to prioritize processing speed for mobile environments. The model was trained on large scale datasets, such as CarDD,

featuring diverse vehicle types under varied lighting and camera angles to simulate real-world road accidents. The results showed that YOLOv8 excels in real time inference speed while maintaining high precision.

Additionally, Hasan et al. (2025) [22] conducted a systematic literature review that analyzed over a decade of AI-driven research to evaluate the trade-offs between speed and accuracy. Their findings concluded that while YOLO based models are optimal for real-time mobile tasks, Instance Segmentation remains the industry standard for insurance grade applications that require exact documentation of the physical damage area. Their analysis also suggested that future systems should integrate image enhancement techniques to mitigate environmental factors like fog or low light.

3.2 **Damage Severity Estimation**:

The second phase involves classifying the intensity of the damage into categories such as minor, moderate, or severe. Qaddour and Siddiqa (2023) [9] introduced a robust estimator utilizing the Inception-ResNetV2 architecture. Their methodology relied on Transfer Learning to train the hybrid model which combines residual connections to prevent data loss with inception modules for complex feature extraction to distinguish between subtle damage intensities across different car parts. The model achieved a high accuracy rate of 92%, validating that deep learning can effectively reduce the subjectivity associated with human assessment.

Complementing this, Mallios et al. (2023) [23] addressed the challenges of "In-The-Wild" images captured by customers in unconstrained environments. They implemented a validation pipeline to ensure image quality before the analysis began. Their technical methodology focused on calculating a "Damage Ratio" by computing the Intersection over Union (IoU) between the damage mask and the vehicle part mask. This quantitative approach allows the system to determine the severity based on the actual percentage of deformation. Their results proved that this method provides a more objective standard for deciding whether a part requires repair or replacement compared to traditional visual estimations.

Commented [ب1]: This can help us if we can do the same thing they did

3.3 **Repair** Cost Estimation:

The final technical phase involves converting visual findings into standardized financial valuations. Elbhrawy et al. (2024) [24] developed the Cost Estimation System (CES) that does not predict price directly via AI but uses a Rule-Based Calculation Engine that integrates AI-detected damage (part + severity) with a pricing database and structured customer data (vehicle make, model, year of manufacture) to calculate the total using explicit formulas:

• Cost of Parts (Cp) = Σ(price of each part)

• Labor Cost (Cl) = R × T (hourly rate × time)

• Overhead Cost (Co) = O × Cl (overhead % × labor cost)

• Profit (Cpr) = P × (Cp + Cl + Co)

• Total Cost (Ct) = Cp + Cl + Co + Cpr

Relying on the visual inputs from the YOLOv8 model (78.50% precision) and Transformer-based body part detection, the system retrieves correct part prices from a database and calculates total cost through transparent mathematical formulas rather than black-box machine learning predictions. The CES successfully generated comprehensive itemized repair reports including parts, labor, and painting costs, providing satisfactory performance for automating claims workflows while maintaining full traceability and regulatory compliance. This approach offers explainability advantages over predictive models, allowing insurance adjusters and customers to understand exactly how repair costs are calculated, though it requires maintaining accurate pricing databases and updating labor rates regularly.

3.4 Discussion and Summary:

Overall, the reviewed studies demonstrate that automated vehicle damage assessment is most effective when implemented as a multi-stage pipeline rather than a single predictive model. Detection models such as Mask R-CNN provide highly accurate damage detection suitable for documentation level analysis, while lightweight detectors like YOLO offer practical real-time performance required for mobile applications. For severity estimation, deep convolutional networks using transfer learning significantly provide consistent classification outcomes. Finally, cost estimation research indicates that rule-based calculation frameworks combined with AI outputs offer higher transparency and reliability compared to purely predictive pricing models.

3.5 Competitive Product Analysis

Automated vehicle damage assessment has been addressed by both global artificial intelligence companies and locally regulated assessment services. To position the proposed system within the current technological and operational landscape, four representative solutions were analyzed: Tractable AI, Inspektlabs, Ravin AI, and Taqdeer. These systems differ in target users, operational models, and technological approaches, providing a comprehensive basis for comparison.

- **Tractable AI:**

Tractable AI is a global technology company specializing in AI-driven claims automation for insurance providers. The platform primarily serves insurance companies and large enterprise claim departments rather than individual vehicle owners. Its core value proposition centers on accelerating claims processing through automated visual assessment.

Publicly available information indicates that Tractable employs deep learning-based computer vision models to analyze vehicle images and generate damage assessments. The system performs image-based damage detection and produces repair cost estimates integrated into insurer workflows. While specific datasets used for model training are not publicly disclosed, Tractable reports training its models on large-scale proprietary insurance datasets to achieve enterprise-level performance.

The platform emphasizes end-to-end automation, including damage detection, cost estimation, and workflow integration with insurer systems. However, it operates primarily as a B2B enterprise solution rather than a consumer-facing application [25].

Figure 6: tractable website's homepage

- **Inspektlabs:**

Inspektlabs provides an AI-powered vehicle inspection platform targeting insurance companies, fleet operators, and rental service providers. The system is designed to automate vehicle inspection processes using computer vision techniques.

According to publicly available descriptions, Inspektlabs uses deep learning models for damage detection, automated inspection reporting, and fraud detection. The platform also supports text recognition capabilities such as VIN or license plate extraction, indicating integration of computer vision and optical character recognition (OCR) technologies.

While detailed information regarding specific model architectures or datasets is not publicly disclosed, Inspektlabs highlights its use of AI-based inspection engines trained on diverse vehicle datasets. The system focuses on automating inspection workflows and generating structured inspection reports suitable for enterprise use [26].

Figure 7: inspektlabs website's homepage

- **Ravin AI:**

Ravin AI specializes in automated vehicle inspection and fleet management solutions. The company primarily serves fleet operators, leasing companies, and mobility service providers. Its system focuses on 360-degree vehicle scanning and automated condition reporting.

Ravin AI utilizes computer vision and machine learning techniques to detect visible vehicle damage from image or video input. The platform emphasizes high-throughput inspection and scalability for large vehicle fleets. Public descriptions indicate that Ravin employs AI-driven image analytics for damage detection and condition monitoring; however, detailed technical specifications or dataset information are not disclosed. Unlike insurance-focused systems, Ravin's operational model centers on lifecycle vehicle inspection rather than formal insurance claim settlement processes [27].

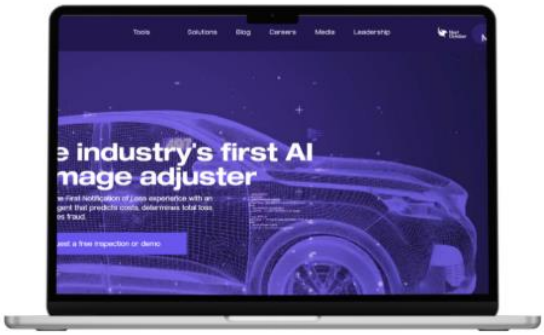

Figure 8: RavinAI website's homepage

- **Taqdeer**

Taqdeer is a Saudi-based vehicle damage assessment service operating under regulated valuation frameworks. Unlike AI-driven global platforms, Taqdeer relies primarily on certified human assessors who conduct physical inspections at authorized centers.

The system produces standardized electronic damage assessment reports used by insurance companies to determine compensation and repair scope. While Taqdeer utilizes centralized electronic reporting systems for documentation and record management, damage detection and cost estimation are performed manually by accredited professionals rather than automated AI models.

As a locally regulated system, Taqdeer is fully integrated into the Saudi insurance ecosystem and serves individual vehicle owners directly. Its operational model emphasizes regulatory compliance and standardized valuation procedures [7].

Figure 9:tagdeer website's homepage

3.5.1 Comparison with CrashLens

CrashLens differentiates itself by targeting individual users within the Saudi market through a mobile-first, localized solution.

Feature	Taqdeer	Tractable AI	Inspektlabs	RavinAI	CrashLens
Automated Damage Detection	✕	✓	✓	✓	✓
Damage localization (bounding boxes / segmentation)	✕	✓	✓	✓	✓
Automated Severity Classification	✕	✓	✓	✓	✓
Automated Repair Cost Estimation	✕	✓	✓	✓	✓
Structured Digital Report Generation	✓	✓	✓	✓	✓

Real-Time Image Capture Guidance	✕	✓	✓	✓	✓
OCR-Based Vehicle Data Extraction	✕	✓	✓	✓	✓
User Objection Submission System	✓	✕	✕	✕	✓
Administrative Case Review & Approval	✓	✓	✓	✓	✓
Consumer-Focused (B2C) Accessibility	✓	✕	✕	✕	✓
Saudi Market Localization	✓	✕	✕	✕	✓
Case Status Tracking	✓	✓	✓	✓	✓
Report Download (PDF / Digital Export)	✓	✓	✓	✓	✓

Table 4: Feature comparison

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CrashLens focuses on improving accessibility to vehicle damage assessment by bridging the gap between enterprise-oriented global solutions and traditional manual assessment services.

3.5.2 Key findings

The comparison table highlights that while global AI platforms provide automated damage detection and cost estimation capabilities, they primarily target enterprise-level insurance workflows and lack consumer-focused localization. Conversely, Taqdeer provides regulatory integration and objection handling but relies on manual inspection. The analysis reveals an opportunity for a localized, consumer-accessible system that integrates automated detection with structured administrative workflows.

4 System Description

The system requirements were selected mainly based on the users 'needs, which were discovered by conducting interviews.

4.1 Users

CrashLens is designed to serve three main types of users Each user category interacts with the system differently based on their role and responsibilities within the damage assessment and verification process.

4.1.1 Individual Users (Vehicle Owners)

Individual users are vehicle owners aged 18 years and above who have experienced a car accident and require a preliminary damage assessment and repair cost estimation. They use the mobile application to capture images of the damaged parts of the vehicle and enter the accident reference number.

Users are expected to have:

- Basic smartphone usage skills
- A general educational background (no specialized knowledge required)
- Basic English proficiency, since the application interface is in English

Their knowledge of insurance procedures may vary; therefore, the system is designed to simplify the assessment process and reduce reliance on manual inspections.

4.1.2 Administrators (Admin)

Administrators are responsible for supervising and managing the overall system through a dedicated dashboard. Their role includes reviewing reports, monitoring system performance, and handling user objections.

They are expected to have:

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- At least an intermediate educational level
- Experience in managing digital platforms or administrative systems

- Background of vehicle damage types and assessment procedures
- Moderate technical literacy
- Basic English proficiency to operate the system effectively

Programming knowledge is not required, but familiarity with dashboards and data review is essential.

4.2 Requirements Elicitation

The requirements for the proposed vehicle damage assessment system were gathered through interviews with domain experts (see appendix A). Interviews were conducted with professional vehicle damage assessors working in certified assessment centers in different branches of taqdeer workshops. This approach ensured the collection of reliable, experience-based insights that accurately reflect real-world assessment workflows and challenges.

Interviews were selected because they provide deeper insights into complex operational procedures and allow experts to describe real-world scenarios that cannot be fully captured through surveys.

The elicitation process focused on identifying system needs, operational constraints, performance expectations, and potential improvement opportunities to support the development of an efficient and intelligent damage assessment platform.

A total of three professional vehicle damage assessors participated in the interviews. All participants actively work in the damage assessment field and possess practical experience Their domain expertise ensured that the elicited requirements were realistic, practical, and aligned with current industry standards.

The elicitation process resulted in a clear understanding of system needs and operational boundaries. The interviews enabled the identification of major functional capabilities, key non-functional expectations, and real-world constraints affecting automated damage assessment.

Additionally, the process clarified important system limitations, such as the inability to detect internal damage without vehicle dismantling.

Overall, the elicitation phase provided a strong foundation for defining precise system requirements and guiding the architecture of the proposed solution.

4.3 Architecture

CrashLens is designed using the MVC (Model–View–Controller) architecture model. This architecture separates the application into three components:

1. **Model:**
   It is the part that deals with data and processing it. This part can be used across multiple views.
2. **View:**
   This part is mainly associated with the User Interface (UI), and it is used to provide the visual representation of the model's data.
3. **Controller:**
   It acts as an intermediary between the model and the view. This part can be used with different views.

The model receives input from the user images, processes them using the AI engine, and stores the results in the database. Then, when the view needs the data, the controller will request it from the model and forward it to the view. Other inputs can come from the view and the controller will pass them to the model that will be stored in the database.

The reason for using this architecture is that it provides clear separation of components which makes it easier to develop and manage the code. It will also provide code reusability. Both reasons are good for cross-platform development.

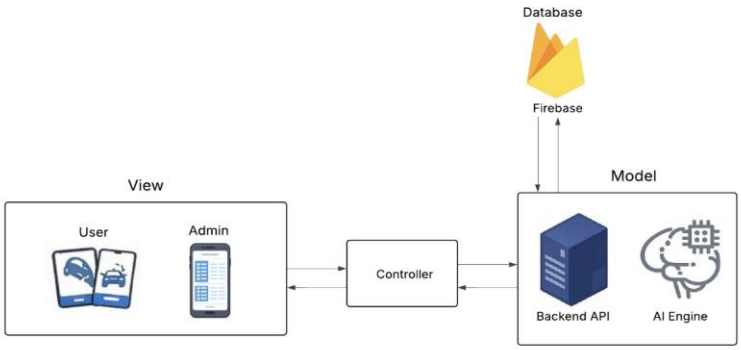

Figure 10: Crashloense high level client server architecture

4.4 Use Case Diagram

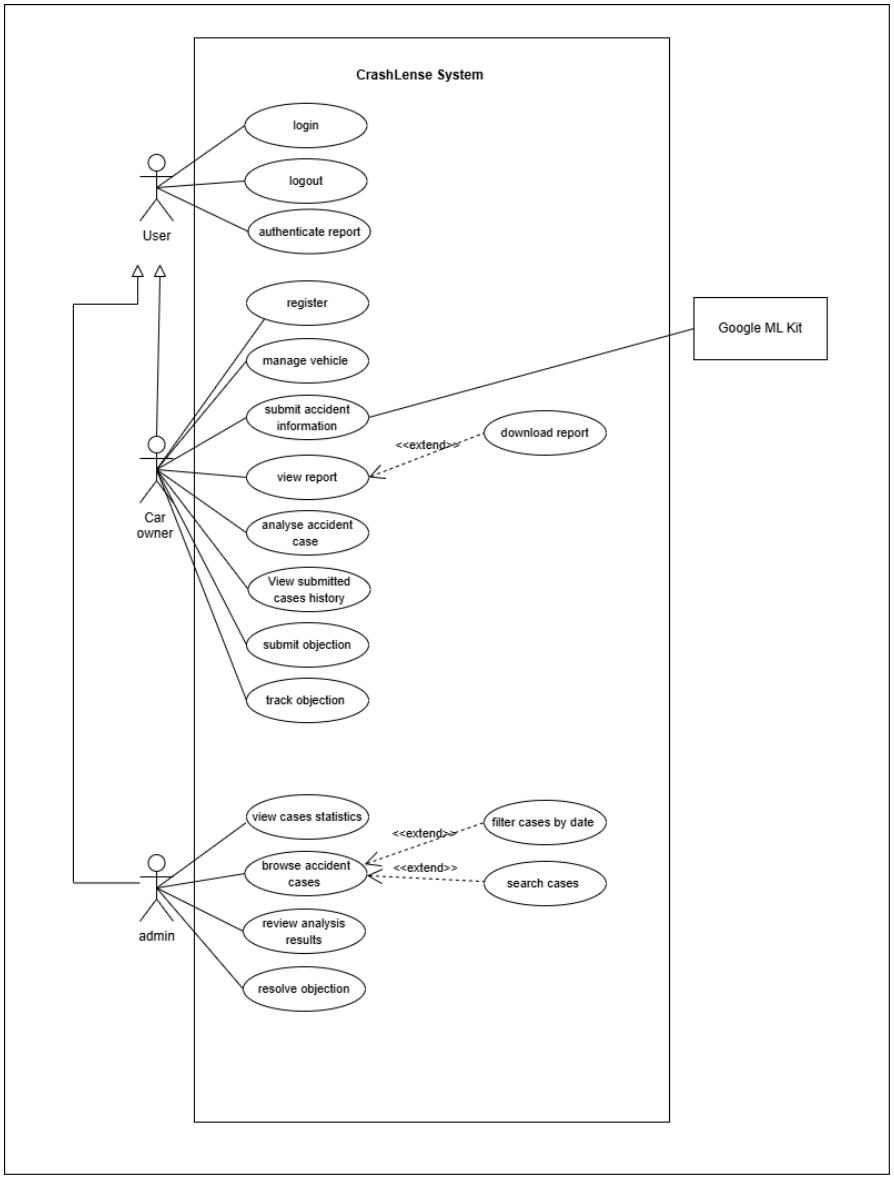

Figure 11: use case diagram of Crashlense

5 Product Backlog

The following Table displays the Definition of Ready table for Crashlense

Example Definition of Ready	
✓	Business value is clearly articulated
✓	Details are sufficiently understood
✓	Dependencies are identified; no blocking dependencies exist
✓	Team is appropriately staffed relative to the PBI
✓	Estimated and small enough to be completed during sprint
✓	Acceptance criteria are clear and testable
✓	Performance criteria, if any, are defined and testable
✓	Team understands how to demo the completed PBI

Table 5: Crashlense Definition of Ready

5.1 Product Backlog Table

The following Table displays the Product Backlog table Crashlense, and it contains the user stories for functional and non-functional requirements.

PBI #	PBI	Size	Type	Acceptance Criteria
1	As a new user, I want to register so that I can access the CrashLens system.	3	Feature	If the user enters a valid phone number, National ID and name then clicks "Continue", then the system should send an OTP to that number. If the user enters the correct OTP within the allowed time, then the system should check whether the phone number already exists.

				If the phone number does not exist, then the system should create a new user account and log the user in automatically. If the account is created successfully, then the user should be redirected to the home page. If the phone number or national ID is invalid or empty, then the system should display a validation message and prevent sending the OTP. If the OTP is incorrect or expired, then the system should display an error message and allow the user to request a new OTP. If the user has not submitted any accident cases, then the system may allow correction of the National ID before completing the first case submission. If the user has already submitted at least one accident case, then the system should prevent modification of the National ID and display an informational message.
2	As a registered user, I want to log into my account so that I can manage my accident cases.	3	Feature	If the user enters a valid phone number and clicks “Continue”, then the system should send an OTP to that number. If the user enters the correct OTP within the allowed time, then the system should check whether the phone number already exists.

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				If the phone number exists, then the system should authenticate the user and log the user in automatically. If the phone number doesn't exist display a message indicating the user to sign up. If login succeeds, then the user should be redirected to the home page. If the phone number is invalid or empty, then the system should display a validation message and prevent sending the OTP. If the OTP is incorrect or expired, then the system should display an error message and allow the user to request a new OTP.
3	As user, I want to log out so that my account remains secure.	1	Feature	If the user clicks the "Log Out" button from any page, then the system should terminate the current session and redirect the user to the login page.
4	As a user, I want to add my vehicle information so that I can register my car in the system and use it later when creating accident cases.	3	Feature	If the user opens the "Add Vehicle" page, then the system should display fields for entering vehicle details (e.g., make, model, year, plate number, and color). If the user fills in all required fields and submits the form, then the vehicle should be saved successfully to the user's account. If any required field is missing, then the system should display validation messages and prevent submission.

				If the vehicle is added successfully, then it should appear in the user's vehicle list and be available for selection when creating accident cases.
5	As a user, I want to scan my vehicle registration to automatically extract and save my vehicle information.	5	Feature	If the user selects "Scan Vehicle Registration", then the camera should open to capture the registration document. If the image is captured clearly, then the system should run OCR to extract vehicle information (e.g., plate number, vehicle type, model year, and color). If the extracted data is detected successfully, then the fields should be auto filled and shown to the user for review. If the OCR fails to detect some fields, then the user should be allowed to edit or manually complete the missing information. If the user confirms the extracted data, then the vehicle should be saved to the user's account.
6	As a user, I want to modify my saved vehicle information so that I can update incorrect or outdated details.	3	Feature	If the user selects a vehicle from their vehicle list and chooses to edit it, then the system should allow modification of only the plate number and color fields, while other vehicle details (e.g., make, model, and year) remain read-only. If the user updates the allowed fields and submits the form, then the updated vehicle information should be saved successfully to the user's account.

				If any required editable field is missing or invalid, then the system should display validation messages and prevent submission. If the modification is completed successfully, then the updated vehicle information should appear immediately in the user's vehicle list. If the user attempts to edit restricted fields, then the system should prevent modification. If there are previous accident cases/reports linked to that vehicle, then those existing cases should keep displaying the vehicle details that were recorded at the time the case was created. If the user views a past report after updating the vehicle, then the report should still show the original recorded vehicle details for that report
7	As a user, I want to delete a saved vehicle from my account so that I can remove vehicles I no longer use.	3	Feature	If the user selects a vehicle from their vehicle list and chooses the delete option, then the system should display a confirmation message before deletion. If the user confirms the deletion, then the vehicle should no longer appear in the user's vehicle list but should be archived rather than permanently deleted. If the vehicle is archived, then the system must retain its information, so it

Commented [لب5]: Check , the idea is that if a case has been submitted the report associated with it should keep a copy of the vehicle information at that time in a database or something instead of facingforcing the user to delete the car and re add it again

				continues to appear in all related reports.
8	As a user, I want to create and submit my accident case so that it can be analyzed by the system.	3	Feature	If the user clicks the “Create New Case” button, then the system should open a new accident case form. If the user completes all required information and uploads at least one image and at most 15 images then presses the Submit button, the system should send the case data and images to the server for analysis. If the submission is successful, then the user should see a confirmation message and the case status should change to “Under Analysis.”
9	As a user, I want to capture vehicle accident damage photos using my camera so that the system can analyze the accident.	5	Feature	If the user selects “Capture Damage Photos”, then the system should open the camera to take images of the vehicle. If the photo is clear and captured successfully, then it should be uploaded and attached to the accident case. If the image quality is poor (e.g., blurry or too dark), then the system should prompt the user to retake the photo. If multiple damages are present, then the system should allow capturing several images to cover all damaged areas. If the user finishes capturing images, then the system should allow them to proceed to the analysis step.
10	As a user, I want to remove an uploaded image before submission	2	Feature	If the user views the uploaded images before submission, then each image should display a “Remove” option.

	so that I can correct mistakes.			If the user selects the “Remove” option on an image, then the image should be deleted from the case preview. If the user removes all images, then the system should prevent submission until at least one image is uploaded.
11	As a user, I want guidance while capturing photos so that I can submit clear and valid images.	5	Feature	If the user opens the camera to capture photos, then the system should display on-screen guidance. If the lighting is too low or too bright, then the system should show a warning asking the user to adjust lighting. If the image is blurry, then the system should prompt the user to retake the photo. If the vehicle is not properly positioned in the frame, then the system should display visual outlines to help align the car.
12	As an ML engineer, I want to train a YOLO-based model so that the system can detect vehicle damage.	8	Technical Work	Model is trained. Validation metrics are recorded. Model achieves acceptable mAP threshold.
13	As a user, I want the system to detect damaged areas in my submitted images so that I can visually understand the affected parts.	8	Feature	If the user submits an accident case with vehicle images, then the system should analyze the images to detect damaged areas. If damaged areas are detected, then the system should highlight the damage visually (e.g., using bounding boxes).

				If multiple damaged areas are present, then the system should detect and display all identified regions.
14	As an ML engineer, I want to train a model to classify damage severity so that the system can estimate impact level.	8	Technical Work	Severity categories (e.g., minor, moderate, severe) are defined. Mapping logic between detection output and severity is documented. Threshold values are specified. Model is trained and validated. Performance metrics are recorded. Model reaches acceptable accuracy threshold.
15	As a user, I want to see the severity level of my vehicle damage so that I understand the seriousness of the accident.	3	Feature	If the analysis is completed, then the system should display the damage severity level (minor, moderate, or severe) in the case results.
16	As a user, I want to see the estimated repair cost so that I can understand the expected financial impact.	3	Feature	If the analysis is completed, then the system should display the estimated repair cost range in the case results. If the case is still under analysis, then the cost estimate should not be shown and the status should indicate processing.
17	As a user, I want a structured report generated after analysis so that I can review my accident assessment clearly.	5	Feature	If the analysis is completed for an accident case, then the system should automatically generate a structured report for that case. If the report is generated, then it should include the user's basic information, vehicle information, uploaded images, detected damaged areas, severity

				classification (minor, moderate, severe), and the estimated cost range. If the report includes visual results, then it should display the highlighted damaged regions (e.g., bounding boxes or masks) for clarity. If the user opens the report, then the report should be viewable inside the application. If the report is generated successfully, then it should be saved and linked to the accident case for future access.
18	As a user, I want to access my previous reports so that I can review past accident cases.	3	Feature	If the user opens the Case History section, then the system should display a list of all previously submitted accident cases with their report status. If the user selects a specific case, then the system should open the associated report and display the saved assessment results. If a case is still under analysis, then the system should show the current status and indicate that the report is not yet available. If the user has no previous cases, then the system should display a clear message indicating that no reports are available.
19	As a user, I want to download my assessment report as a PDF so that I can share it if needed.	3	Feature	If the user opens an assessment report, then the system should provide an option to download the report as a PDF file.

				If the user selects the Download PDF option, then the system should generate and download the report in PDF format to the device. If the report generation is successful, then the downloaded file should contain all report details and visual results.
20	As a third party, I want to verify the accident report through a QR code so that I can confirm its authenticity and accuracy.	5	Feature	If the third party scans the QR code included in the report, then the system should open a secure verification page for that report. If the report exists and is valid, then the system should display the report details.
21	As an admin, I want to browse submitted accident cases so that I can manage incoming assessments.	3	Feature	If the admin opens the Cases Management section, then the system should display a list of all submitted accident cases. If cases exist, then each case should show basic information (case ID, user, date, and current status). If the admin selects a specific case, then the system should open the full case details and associated images. If there are no submitted cases, then the system should display a message indicating that no cases are available.
22	As an admin, I want to filter accident cases by date so that I can quickly locate cases within a specific time range.	3	Feature	If the admin opens the Cases Management page, then the system should provide date filter fields. If the admin selects a date range, then the system should display only cases submitted within that period.

				If the admin clears the date filter, then the system should display all cases again.
23	As an admin, I want to search for a case so that I can directly access related cases.	3	Feature	If the admin enters a valid vehicle plate number or case ID in the search field, then the system should display all cases associated with that plate number. If no cases exist for the entered vehicle plate number or case ID, then the system should show a message indicating that no matching cases were found. If the admin clears the search input, then the system should return to displaying the full list of cases.
24	As an admin, I want to view case statistics so that I can monitor system performance and case trends.	5	Feature	If the admin opens the Statistics Dashboard, then the system should display summarized case statistics (e.g., total cases, approved cases, under analysis). If data exists, then the system should present visual charts showing trends over time (e.g., cases per day/month). If the dashboard loads successfully, then all statistics should reflect the latest system data.
25	As an admin, I want to review a case to ensure the assessment is accurate before finalization.	3	Feature	If the admin opens the Cases page and selects a specific case, then the system should display the full case details and the generated report. If the admin reviews the report, then they should have the option to either modify the report or approve the case. If the admin chooses to modify the report, then the system should allow

				editing of the assessment details and save the updated report. If the admin approves the case, then the system should mark the case status as Approved. If the admin completes either action, then the system should update the case status in the user's account.
26	As a user, I want to submit an objection if I disagree with the result so that my case can be re-evaluated.	3	Feature	If the user opens the Objections page, then the system should display a list of eligible cases for objection. If the user selects a case, then the system should allow them to enter the objection reason. If the user submits the objection successfully, then the case status should change to Under Review.
27	As a user, I want to track the status of my submitted objection so that I can stay informed about its review progress.	3	Feature	If the user opens the Objections section, then the system should display a list of their submitted objections with their current status. If an objection is under review, then the system should display the status as Under Review. If the objection is accepted, then the system should update the status to Accepted. If the objection is rejected, then the system should update the status to Rejected. If the admin updates the objection decision, then the status should be

				updated immediately in the user's account. If the objection status changes, then the system should send a push notification to the user indicating that the objection status has been updated.
28	As an admin, I want to review user objections so that I can decide whether reassessment is needed.	5	Feature	If the admin opens the Objections page, then the system should display a list of submitted user objections. If the admin selects a specific objection, then the system should show the related case details and objection reason. If the admin accepts the objection, then the system should allow the admin to modify the report and update the case status to Updated. If the admin rejects the objection, then the system should keep the report unchanged and mark the objection as rejected. If a decision is made, then the system should update the case status in the user's account.
29	As a first-time user I want the app to be simple so that I would be able to complete the accident case process within 5 minutes under normal operating conditions.	3	Feature	If a first-time user follows the normal accident case submission workflow under standard network conditions, then the complete process shall not exceed 5 minutes from case creation to final submission. If required information or images are missing during submission, then the system shall immediately display clear validation messages within 5 seconds. The submission process should not require unnecessary steps

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30	As a user, I want the system to be available at least 99% of the time so that I can access my accident reports whenever needed.	3	Feature	If a user attempts to access the system, then the system should be operational and accessible 99% of the time measured monthly. If scheduled maintenance occurs, then users should be notified in advance. If downtime exceeds the allowed threshold, then the event should be logged and reported.
31	As a user, I want the application pages to load within 10 seconds from the moment I initiate a request to access any page, so that I can access all functions without delays.	3	Feature	If user open any page, then the page should fully load within 10seconds under normal network conditions on supported mobile devices.
32	As a user, I want the app interface to adapt to different screen sizes so that I can complete tasks without layout issues.	3	Feature	If the app is opened on different supported screen sizes, then the UI should scale and display all essential elements without overlapping, truncation, or unreadable text. If a screen becomes too small for a specific layout element, then the system should apply responsive behaviour (e.g., stacking components vertically) rather than hiding required actions.
33	As a user, I want the application to show helpful error messages and handle unexpected errors smoothly so that I can easily understand and resolve issues.	2	Feature	If the user performs an action with missing or invalid input, then the system should display a clear and understandable error message explaining the issue.

Table 6: product backlog table of Crashlense

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7 Appendix A:

7.1 Interview 1:

Q1: What is the current damage assessment procedure?

the process begins at the reception counter, where vehicle data is registered in the system, including the accident number and vehicle registration details. Once registered, the vehicle proceeds to the assessment lane. The assessor conducts a standardized inspection that includes capturing photographs from all angles of the vehicle, photographing visible damage areas, documenting the seatbelt to verify its functionality, capturing the dashboard and odometer, and recording the chassis number. This documentation process is mandatory for all cases and ensures consistency across assessments.

Q2: How are hidden or internal damages handled?

only visible damage is evaluated during the initial inspection. Internal damage is not examined at this stage because dismantling the vehicle is not performed at the assessment center. If internal damage is later discovered at a workshop, the customer may submit an objection or additional claim. Supporting photos must be provided, and in some cases, another assessor may be assigned to review the additional damage. The system does not approve part replacements without documented evidence.

Q3: How is repair cost calculated after inspection?

repair cost calculation depends on whether the damaged component requires replacement or repair. In cases requiring replacement, the damaged parts are recorded, labor cost is determined by the assessor, and the request is sent to a contracted supplier who provides the spare part price. The total cost is then communicated to the client. If the part requires repair only, the assessor calculates an average repair cost based on market standards.

Q4: What factors affect repair cost?

Several factors influence the final repair cost. These include the type of paint used on the vehicle, as certain finishes such as pearlescent or metallic paints increase cost. The vehicle's age is also significant; vehicles older than ten years may be handled under a different evaluation mechanism. Additionally, whether spare parts are original or commercial affects pricing, while labor cost generally remains standardized.

Q5: How are spare part prices updated?

Spare part prices are determined through contracted suppliers affiliated with the system. Approximately 20–30 suppliers are involved. The suppliers are responsible for setting and updating pricing, and they coordinate with authorized dealerships. The assessment authority does not directly determine spare part prices.

Q6: What challenges do assessors face?

identifying chassis damage is a critical challenge. Assessing mechanical components and determining whether damage is new or pre-existing can also be difficult. In some situations, the assessor must explain to customers that certain damages are unrelated to the current accident. identifying chassis damage is a critical challenge. Assessing mechanical components and determining whether damage is new or pre-existing can also be difficult. In some situations, the assessor must explain to customers that certain damages are unrelated to the current accident.

Q7: What is the average time required for assessment?

Minor damage cases typically take between 10 to 15 minutes. However, major damage cases may require extended review and, in some instances, committee evaluation that can take one to two days.

Q8: What types of damage are most difficult to evaluate?

Electrical damage and mechanical damage were identified as the most difficult to evaluate, especially when multiple components are involved. There is also a risk of overlooking hidden or interconnected damage.

Q9: Are there any suggested improvements?

the system has significantly improved over time and currently operates effectively. No major improvements were suggested.

Q10: How was electronic assessment implemented during Hajj?

During Hajj season, the centre operated continuously due to increased case volume. Electronic and remote assessment was primarily used for minor and clearly visible damage. Major collisions were not evaluated remotely and required in-person assessment or referral to a market valuation expert.

7.2 Interview 2:

Q1: What is the current mechanism used for vehicle damage assessment?

The current mechanism is fully electronic and operates through the Taqdeer system. Upon vehicle arrival, the case is registered digitally and assigned to an assessor. The system guides the assessor regarding required photographs, including general vehicle images, damage areas, structural components, license plate, odometer, airbags, and seatbelts. All damaged parts are recorded in the system, after which labor cost is determined by the assessor and spare part pricing is requested from suppliers.

Q2: How are hidden damages handled?

Internal damages are not identified during the initial assessment. The vehicle must undergo workshop dismantling to detect such damage. If additional issues are discovered, a supplementary claim can be submitted along with supporting images. An assessor may review the updated information before approval.

Q3: How is repair cost calculated?

Repair cost consists of supplier-determined spare part pricing and assessor-determined labour cost. If prior compensation has been issued, the system adjusts compensation accordingly. Additional labour charges are applied if further damage is discovered after dismantling.

Q4: What are the main factors affecting repair cost?

Repair cost is influenced by whether parts are original or commercial, previous repair history, paint type, and whether specialized programming is required for luxury vehicles. Vehicle age may also influence evaluation method.

Q5: How are spare parts priced and updated?

Spare parts are priced by suppliers in coordination with authorized dealers. Pricing responsibility lies entirely with suppliers, not with the assessment authority.

Q6: What challenges do assessors face?

Major challenges include deployed airbags, damaged onboard computer systems, extensive structural damage, and cases requiring underbody dismantling. Diagnostic tools may be required to detect full damage extent.

Q7: How long does assessment take?

Assessment duration varies. Minor cases take approximately 10–15 minutes, moderate cases up to 30 minutes, and heavily damaged vehicles require longer evaluation.

Q8: Which types of damage are hardest to evaluate?

Severe and heavily impacted damages, particularly structural or electronic system damage, are the most difficult.

Q9: What improvements are needed?

I am satisfied with the current system and do not recommend specific improvements.

Q10: How was electronic assessment implemented during Hajj?

The electronic system was applied in certain regions during Hajj, primarily for minor damage cases. It helped manage high case volumes efficiently.

7.3 Interview 3:

Q1: What is the current damage assessment procedure?

the process begins by verifying the accident number, whether issued by Traffic Police or Najm, along with confirming the vehicle registration and customer contact information. Once the vehicle is registered, the assessor conducts a full 360-degree inspection. This includes documenting visible damage through photographs and verifying key vehicle information. The assessor ensures that all required documentation is captured before proceeding with evaluation. The case is then processed within the Najm and Taqdeer systems, and repair costs are calculated based on average workshop market rates rather than dealership pricing, even for newly manufactured vehicles.

Q2: How are hidden or internal damages handled?

only visible damages are assessed during the initial inspection. If hidden or internal damage is suspected, further inspection is required. In such cases, components may need to be dismantled at a workshop to determine the full extent of the damage. Any additional damage discovered after dismantling must be documented and submitted for supplementary review. Approval for additional damage may require supporting evidence and further evaluation.

Q3: How is the repair cost calculated after inspection?

Repair cost is calculated based on several structured criteria. The assessor first classifies the severity of damage as minor, medium, or major. Spare part prices are obtained from approved suppliers affiliated with the Saudi Authority for Accredited Valuers, while labour cost is determined based on average workshop rates. Vehicle category and paint specifications are also considered. Dealership pricing is not used in the calculation. If the repair cost exceeds a defined threshold, such as 50% of the vehicle's market value, the vehicle may be evaluated for total loss.

Q4: What factors affect repair cost?

repair cost is influenced by the severity of damage, the type of damaged components, spare part pricing, and labor cost averages. The vehicle's age plays a role, particularly for vehicles older than ten years. Structural or safety-related damage, such as chassis impact, airbag deployment, or engine damage, significantly increases cost. Additionally, if parts were previously repaired or compensated for, this affects the evaluation outcome.

Q5: How are spare part prices updated?

Spare part prices are provided by approved suppliers affiliated with the Saudi Authority for Accredited Valuers. Pricing is based on the specific part name and the vehicle's chassis number to ensure accuracy. Suppliers coordinate with authorized dealerships when necessary, and they are responsible for maintaining updated and accurate pricing. The assessment authority does not directly set spare part prices.

Q6: What challenges do assessors face?

Major collisions involving structural damage are difficult to evaluate. Deployed airbags, engine damage, and damage affecting safety systems require careful inspection. Cases involving extensive dismantling or hidden damage extension also increase complexity. Additionally, ensuring accurate documentation and detecting potential fraud are ongoing challenges. Dependency on external systems such as Najm reports can also affect workflow efficiency.

Q7: What is the average time required for assessment?

the average inspection time is approximately 15 minutes for standard cases. However, assessment duration varies depending on damage severity. Major or complex cases may require extended evaluation time.

Q8: What are the most difficult types of damage?

The most difficult cases involve severe structural damage, chassis impact, airbag deployment, engine damage, and damage affecting safety-related systems. Total loss evaluations are also complex, particularly when multiple criteria must be assessed. A vehicle may be declared a total loss if repair cost exceeds 50% of its value, if it is older than ten years, if there is no official dealer in Saudi Arabia, or if it is classified as a classic or inherited vehicle.

Q9: Are there any suggested improvements?

the authority continuously updates and improves the system. Suggestions for enhancement included strengthening fraud detection alerts, improving integration between Najm and Bashar systems, enhancing electronic documentation features, and ensuring standardized pricing accuracy across suppliers.

Q10: How was electronic assessment implemented during Hajj season?

During Hajj season, electronic assessment through Najm's remote estimation feature was primarily used for minor damage cases. Remote estimation covered labor costs only and was not applicable for medium or major damage cases. If Najm could not assess the case remotely, it was transferred to the Traffic Department for further processing. The electronic system contributed to managing the high case volume during peak seasonal periods.